Module 10: Linux Exploitation and Privilege Escalation

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# Exercise 08: Identifying Misconfigurations for Privilege Escalation using AI
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Scenario

AI-Assisted Linux Misconfiguration Auditing & Automated File Permission Scanning leverages LLM prompt synthesis to automate SSH connection establishing, remote command execution, and file system auditing across target Linux systems.

Key technical concepts in AI privilege escalation auditing:
1. **Natural Language Audit Script Generation**: Prompting ChatGPT to craft an automated Linux file permissions audit script targeting **192.168.0.51** (**ubuntu** / **toor**):
- SUID binary discovery (`find / -perm -4000`).
- SGID binary discovery (`find / -perm -2000`).
- World-writable file and directory auditing (`find / -perm -2 -ls` / `find / -perm -0002`).
- Exporting results to structured report file **file_permissions_report.txt**.
2. **UNIX Format Conversion (`dos2unix`)**: Converting CRLF line endings on **Privilege_Escalate.sh** via **dos2unix Privilege_Escalate.sh** prior to execution.
3. **Exploitable Binary Identification**: Parsing **file_permissions_report.txt** to identify misconfigured SUID binaries—specifically discovering SUID binary **/usr/bin/cp**—which enables root file overwriting (e.g., overwriting `/etc/passwd` or `/etc/shadow`) for instant privilege escalation.

In this exercise, we access ChatGPT, generate `Privilege_Escalate.sh`, copy the code into **/home/pentester/Scripts/Module 10/Privilege_Escalate.sh**, grant execution permissions (`chmod +x`), sanitize via `dos2unix`, execute the script, open **file_permissions_report.txt**, and identify SUID binary **/usr/bin/cp**.

Objective

The objective of this lab is to help students learn how to:
- Authenticate to ChatGPT and prompt for automated privilege escalation misconfiguration scripts
- Save synthesized script to **/home/pentester/Scripts/Module 10/Privilege_Escalate.sh**
- Grant execution permissions: **chmod +x Privilege_Escalate.sh**
- Convert DOS line endings to UNIX format: **dos2unix Privilege_Escalate.sh**
- Execute automated audit script against target host **192.168.0.51**: **./Privilege_Escalate.sh**
- Audit generated output report **file_permissions_report.txt**
- Locate dangerous SUID binary **/usr/bin/cp** enabling file overwrite privilege escalation

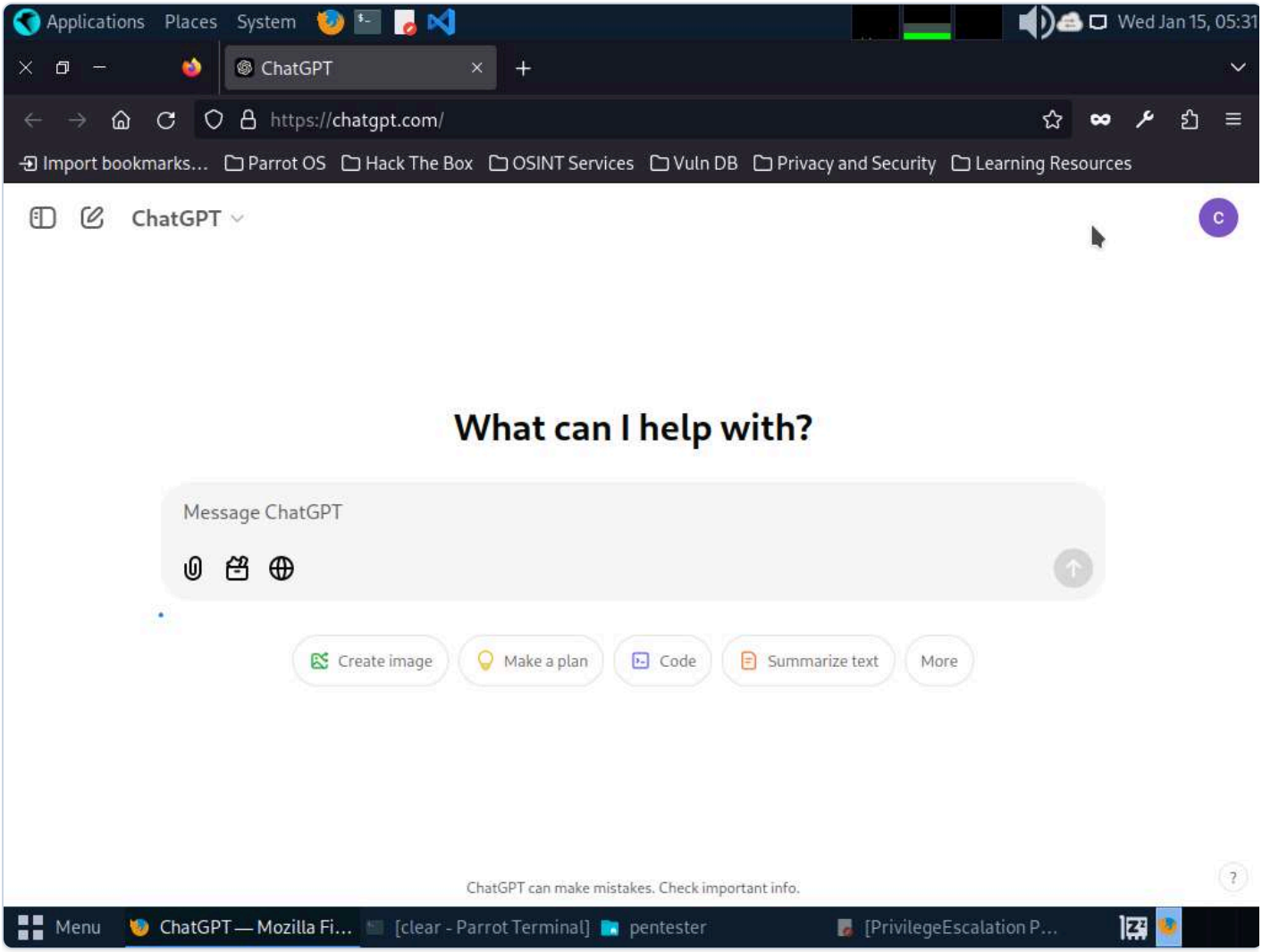
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Task 1: Generate AI Misconfiguration Audit Script, Sanitize Format via dos2unix, Execute Automated Scan, and Audit Findings

In this task, we prompt ChatGPT to generate `Privilege_Escalate.sh`, convert formatting, execute misconfiguration scan against `192.168.0.51`, and inspect `file_permissions_report.txt` for SUID binary `/usr/bin/cp`.

Lab Tasks

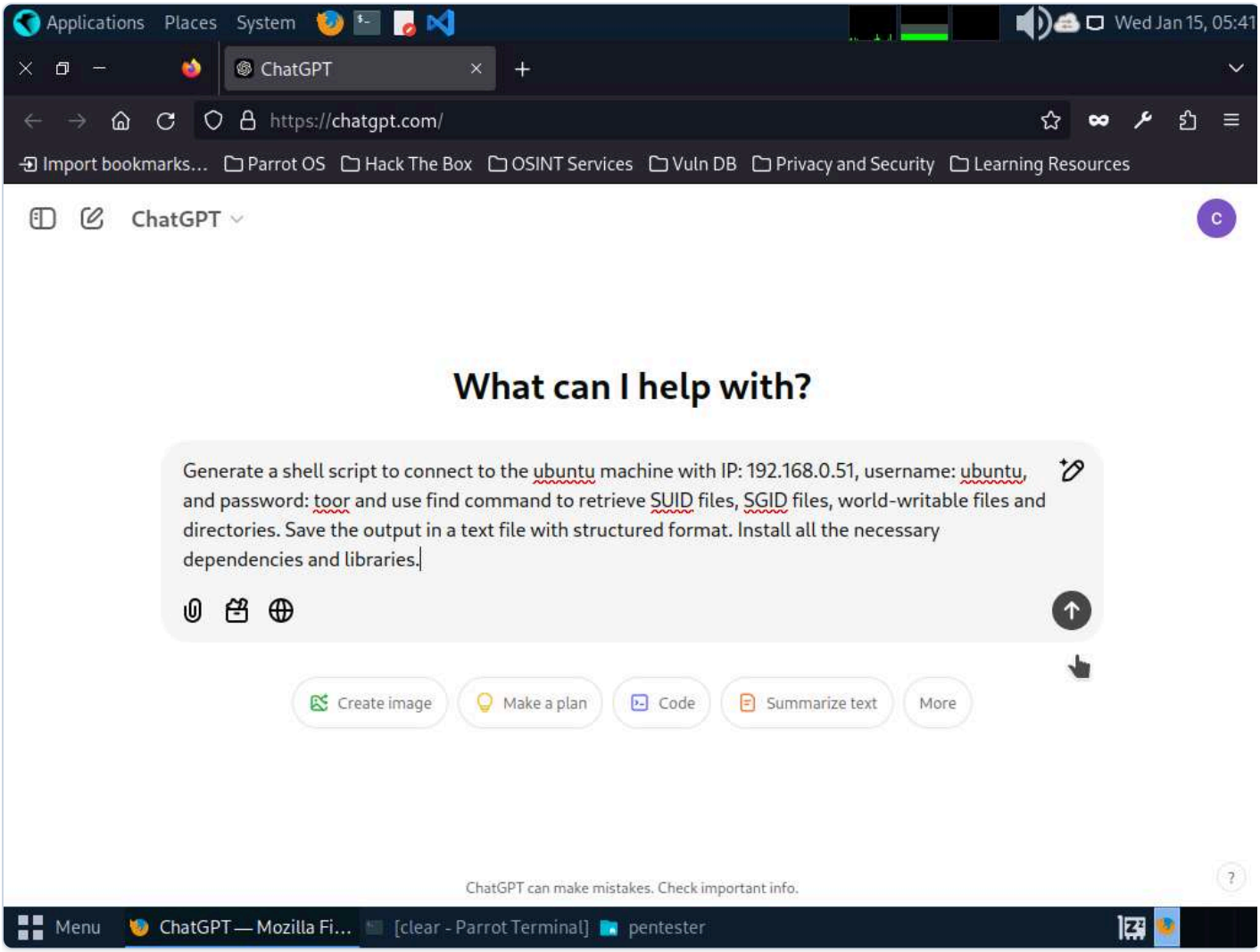
1. Log into **Parrot Security** VM using credentials (**pentester** / **toor**). Open Firefox browser and navigate to ChatGPT (`https://chatgpt.com/`).

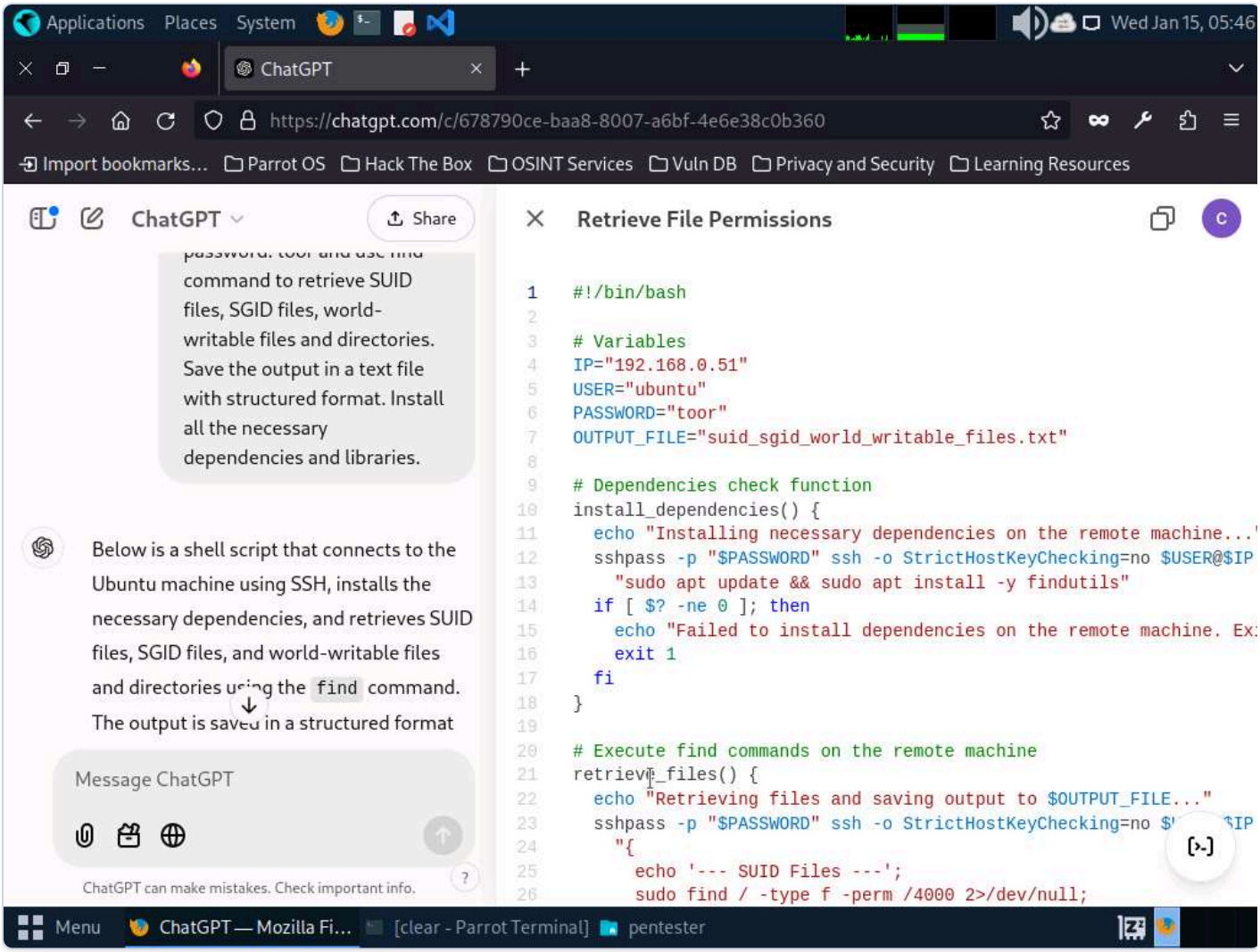


1. Authenticate to ChatGPT using Google credentials.

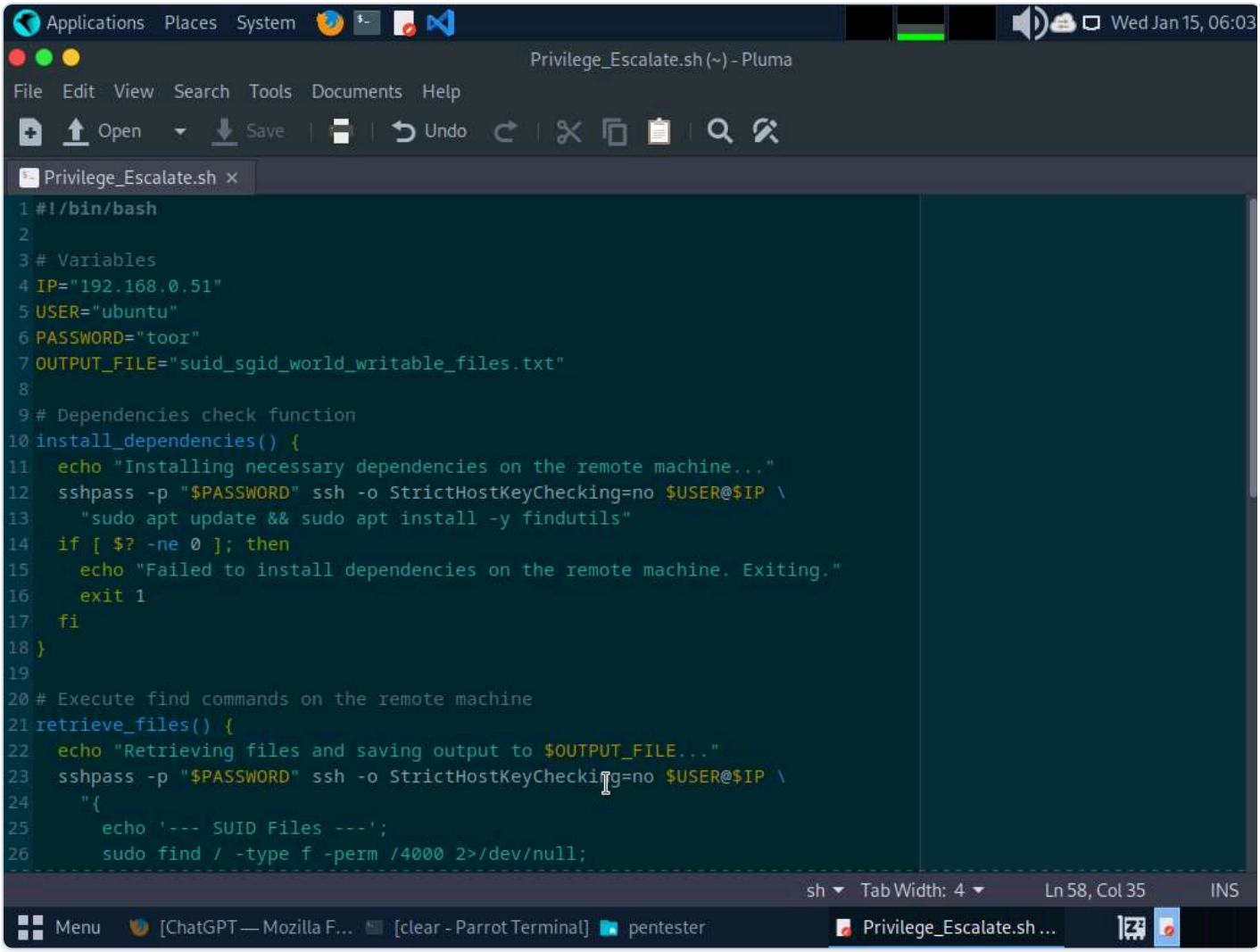
2. Submit the following script generation prompt into ChatGPT:

   Generate a shell script to connect to the Ubuntu machine with IP: 192.168.0.51, username: ubuntu, and password: toor and use the find command to retrieve SUID files, SGID files, world-writable files, and directories. Save the output in a text file with a structured format. Install all the necessary dependencies and libraries.

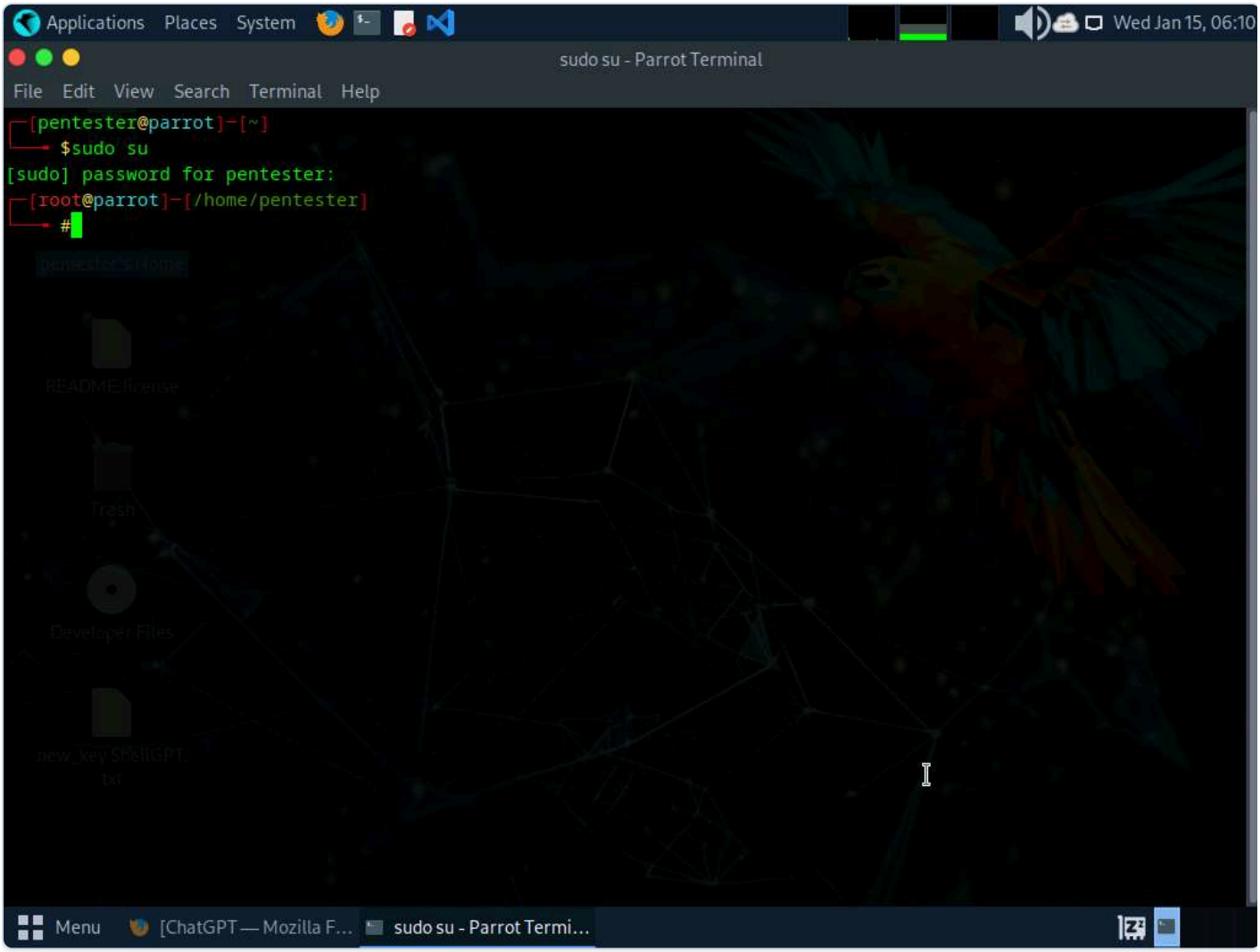

Applications Places System
Wed Jan 15, 05:41
ChatGPT
https://chatgpt.com/
Import bookmarks... Parrot OS Hack The Box OSINT Services Vuln DB Privacy and Security Learning Resources
ChatGPT
C
What can I help with?
Generate a shell script to connect to the ubuntu machine with IP: 192.168.0.51, username: ubuntu, and password: toor and use find command to retrieve SUID files, SGID files, world-writable files and directories. Save the output in a text file with structured format. Install all the necessary dependencies and libraries.
Create image
Make a plan
Code
Summarize text
More
ChatGPT can make mistakes. Check important info.
Menu
ChatGPT — Mozilla Fi...
[clear - Parrot Terminal]
pentester



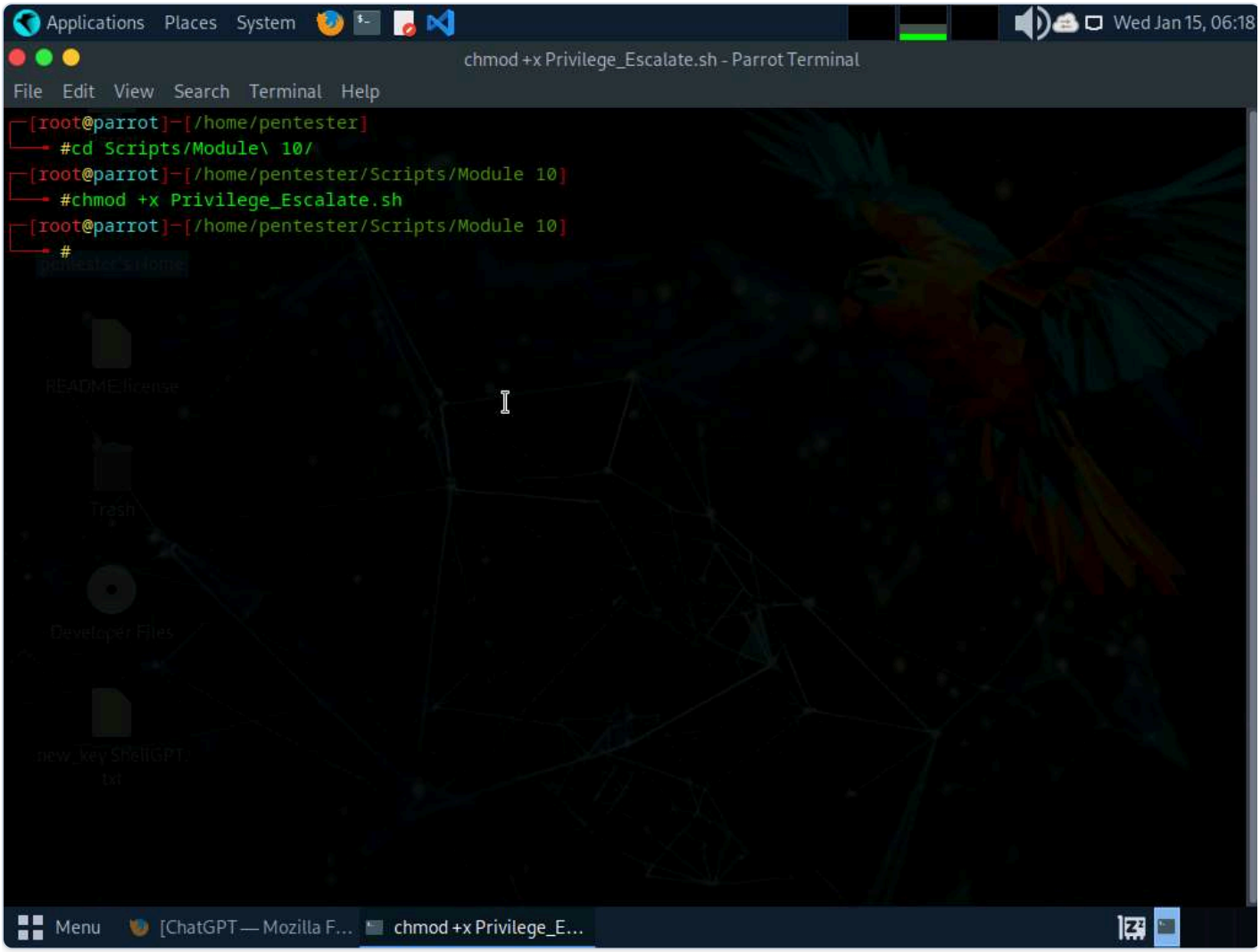
1. Copy the generated shell script code into file **Privilege_Escalate.sh**.



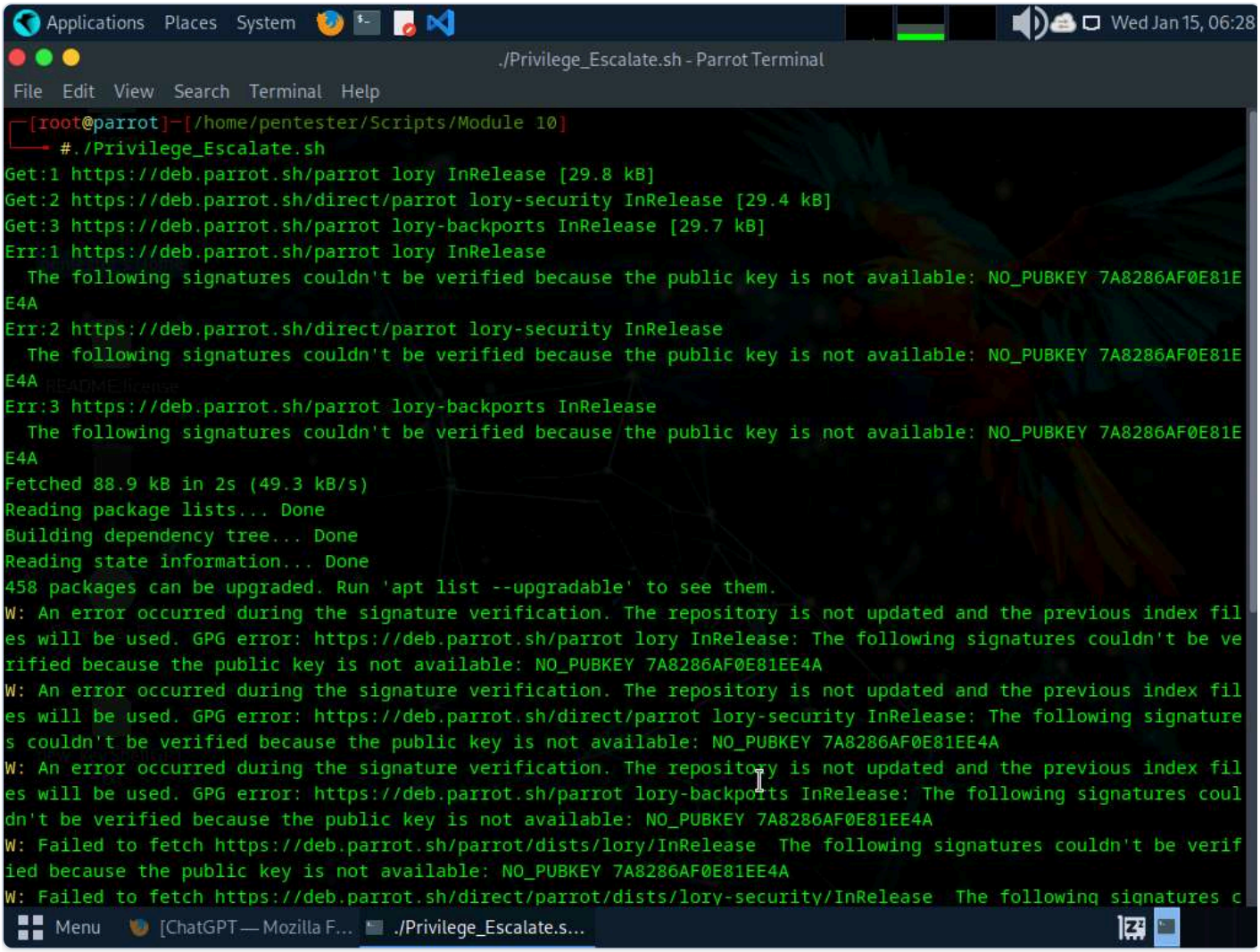
1. Open terminal, obtain root privileges (`sudo su`), and navigate to script directory:
   cd /home/pentester/Scripts/Module\ 10

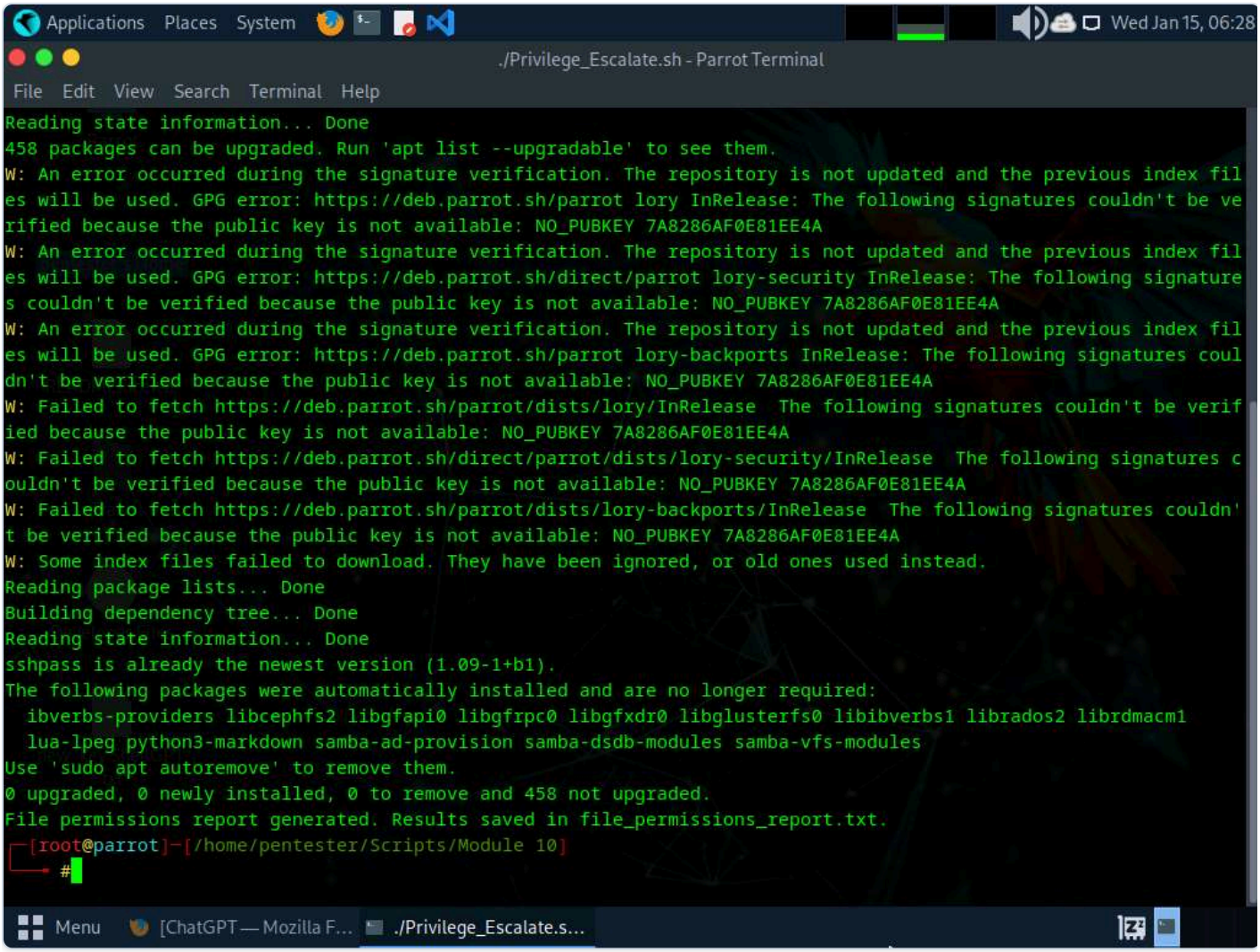


1. Grant executable permissions to script file:
   chmod +x Privilege_Escalate.sh

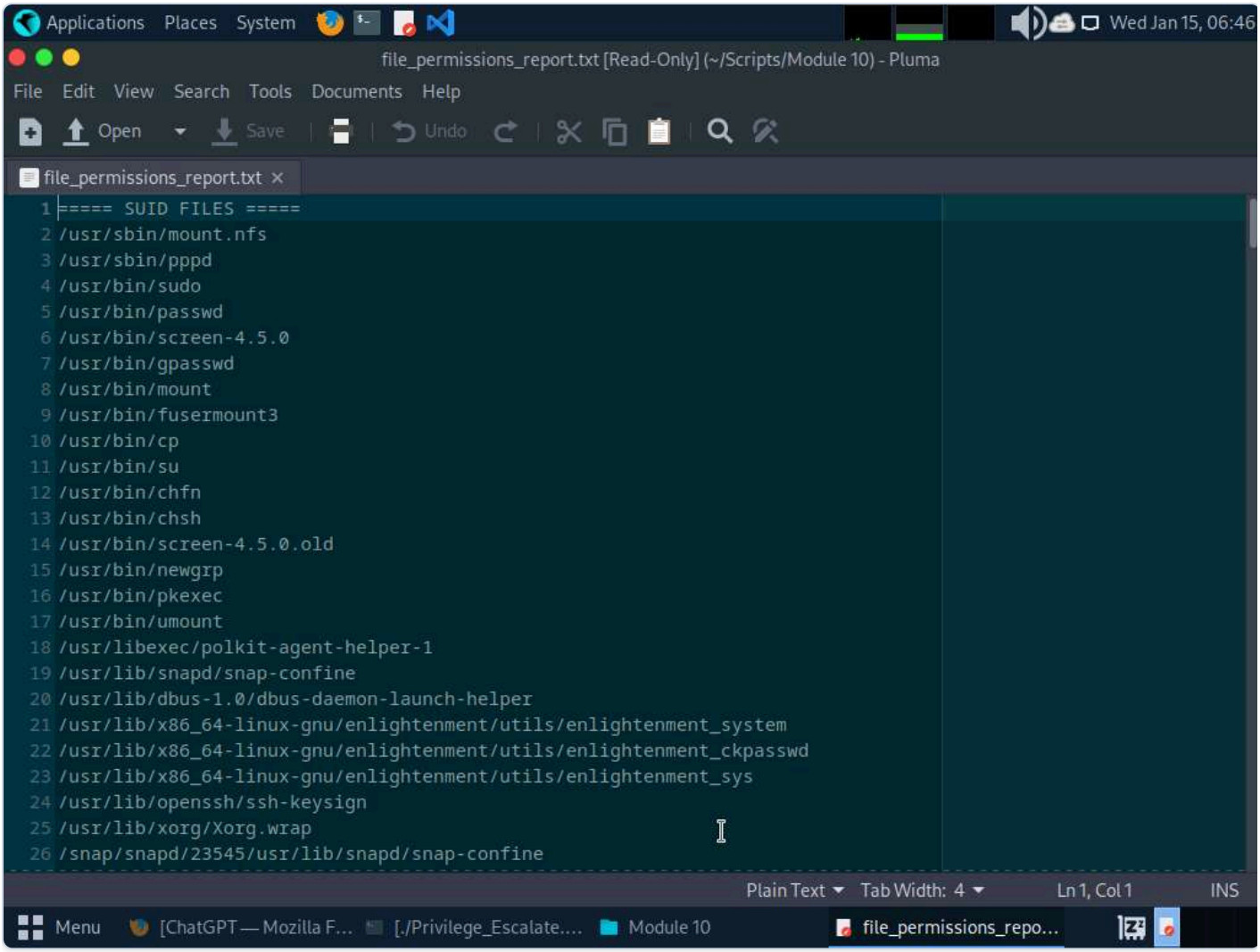


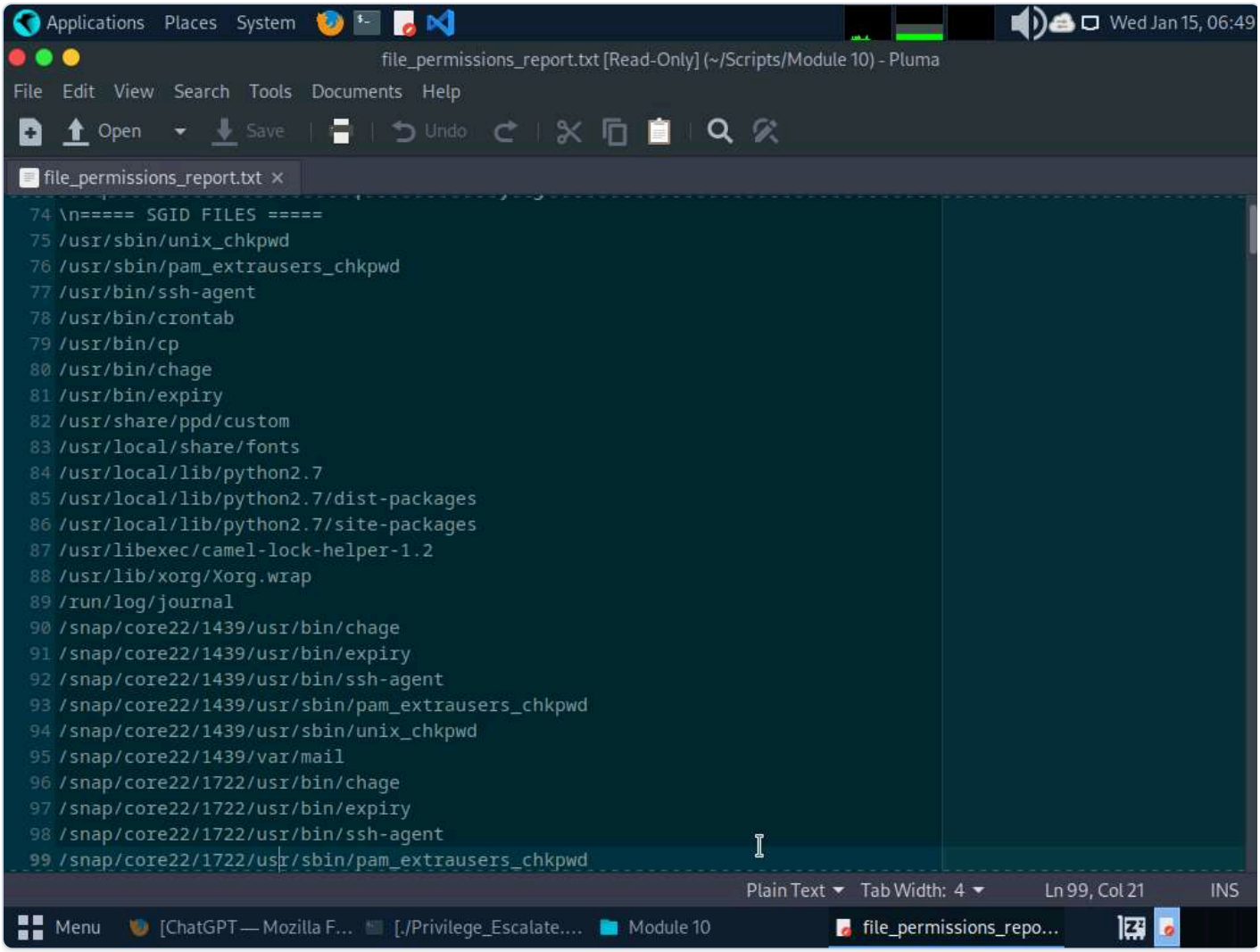
1. Convert script formatting to UNIX line endings and execute misconfiguration audit:
   dos2unix Privilege_Escalate.sh
   ./Privilege_Escalate.sh

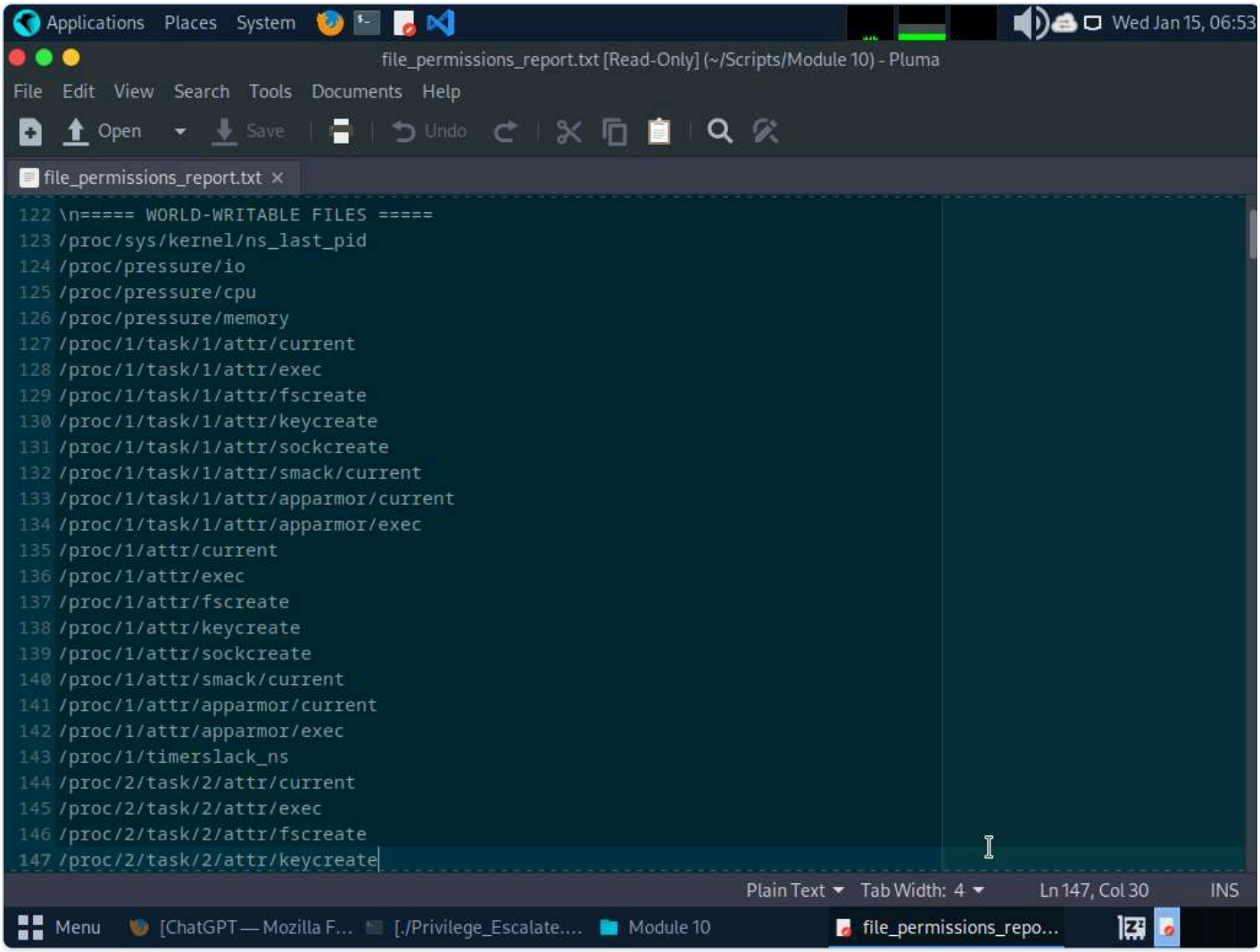

Applications Places System
Wed Jan 15, 06:28
./Privilege_Escalate.sh - Parrot Terminal
File Edit View Search Terminal Help
┌─[root@parrot]─[/home/pentester/Scripts/Module 10]
└──╼ #./Privilege_Escalate.sh
Get:1 https://deb.parrot.sh/parrot lory InRelease [29.8 kB]
Get:2 https://deb.parrot.sh/direct/parrot lory-security InRelease [29.4 kB]
Get:3 https://deb.parrot.sh/parrot lory-backports InRelease [29.7 kB]
Err:1 https://deb.parrot.sh/parrot lory InRelease
The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7A8286AF0E81EE4A
Err:2 https://deb.parrot.sh/direct/parrot lory-security InRelease
The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7A8286AF0E81EE4A
Err:3 https://deb.parrot.sh/parrot lory-backports InRelease
The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7A8286AF0E81EE4A
Fetched 88.9 kB in 2s (49.3 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
458 packages can be upgraded. Run 'apt list --upgradable' to see them.
W: An error occurred during the signature verification. The repository is not updated and the previous index files will be used. GPG error: https://deb.parrot.sh/parrot lory InRelease: The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7A8286AF0E81EE4A
W: An error occurred during the signature verification. The repository is not updated and the previous index files will be used. GPG error: https://deb.parrot.sh/direct/parrot lory-security InRelease: The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7A8286AF0E81EE4A
W: An error occurred during the signature verification. The repository is not updated and the previous index files will be used. GPG error: https://deb.parrot.sh/parrot lory-backports InRelease: The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7A8286AF0E81EE4A
W: Failed to fetch https://deb.parrot.sh/parrot/dists/lory/InRelease The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7A8286AF0E81EE4A
W: Failed to fetch https://deb.parrot.sh/direct/parrot/dists/lory-security/InRelease The following signatures c
Menu
[ChatGPT — Mozilla F...
./Privilege_Escalate.s...



1. Open generated report file **file_permissions_report.txt**:
   cat file_permissions_report.txt
   Observe output: Identifies SUID files, SGID files, world-writable files, and flagged binary **/usr/bin/cp**.


Applications Places System
Wed Jan 15, 06:46
file_permissions_report.txt [Read-Only] (~/Scripts/Module 10) - Pluma
File Edit View Search Tools Documents Help
Open Save Undo
file_permissions_report.txt
1 ===== SUID FILES =====
2 /usr/sbin/mount.nfs
3 /usr/sbin/pppd
4 /usr/bin/sudo
5 /usr/bin/passwd
6 /usr/bin/screen-4.5.0
7 /usr/bin/gpasswd
8 /usr/bin/mount
9 /usr/bin/fusermount3
10 /usr/bin/cp
11 /usr/bin/su
12 /usr/bin/chfn
13 /usr/bin/chsh
14 /usr/bin/screen-4.5.0.old
15 /usr/bin/newgrp
16 /usr/bin/pkexec
17 /usr/bin/umount
18 /usr/libexec/polkit-agent-helper-1
19 /usr/lib/snapd/snap-confine
20 /usr/lib/dbus-1.0/dbus-daemon-launch-helper
21 /usr/lib/x86_64-linux-gnu/enlightenment/utils/enlightenment_system
22 /usr/lib/x86_64-linux-gnu/enlightenment/utils/enlightenment_ckpasswd
23 /usr/lib/x86_64-linux-gnu/enlightenment/utils/enlightenment_sys
24 /usr/lib/openssh/ssh-keysign
25 /usr/lib/xorg/Xorg.wrap
26 /snap/snapd/23545/usr/lib/snapd/snap-confine
Plain Text Tab Width: 4 Ln 1, Col 1 INS
Menu [ChatGPT — Mozilla F... [./Privilege_Escalate.... Module 10 file_permissions_repo...


Applications Places System
Wed Jan 15, 06:49
file_permissions_report.txt [Read-Only] (~/Scripts/Module 10) - Pluma
File Edit View Search Tools Documents Help
Open Save Undo
file_permissions_report.txt
\n===== SGID FILES =====
/usr/sbin/unix_chkpwd
/usr/sbin/pam_extrausers_chkpwd
/usr/bin/ssh-agent
/usr/bin/crontab
/usr/bin/cp
/usr/bin/chage
/usr/bin/expiry
/usr/share/ppd/custom
/usr/local/share/fonts
/usr/local/lib/python2.7
/usr/local/lib/python2.7/dist-packages
/usr/local/lib/python2.7/site-packages
/usr/libexec/camel-lock-helper-1.2
/usr/lib/xorg/Xorg.wrap
/run/log/journal
/snap/core22/1439/usr/bin/chage
/snap/core22/1439/usr/bin/expiry
/snap/core22/1439/usr/bin/ssh-agent
/snap/core22/1439/usr/sbin/pam_extrausers_chkpwd
/snap/core22/1439/usr/sbin/unix_chkpwd
/snap/core22/1439/var/mail
/snap/core22/1722/usr/bin/chage
/snap/core22/1722/usr/bin/expiry
/snap/core22/1722/usr/bin/ssh-agent
/snap/core22/1722/usr/sbin/pam_extrausers_chkpwd
Plain Text Tab Width: 4 Ln 99, Col 21 INS
Menu [ChatGPT — Mozilla F... [./Privilege_Escalate.... Module 10 file_permissions_repo...



This concludes the lab exercise.

> **IMPORTANT:** ❓ **Question 10.8.1.1** On the Parrot Security machine, utilize ChatGPT to generate scripts for identifying misconfigurations that may lead to privilege escalation. Focus on issues such as improper file permissions, misconfigured sudo policies, and exploitable SUID/SGID binaries on the target machine (192.168.0.51). Enter the path of cp file that is obtained by scanning the target machine. [!SUCCESS] ✅ **Answer:** **/usr/bin/cp**